

EDUCATION

- **Cornell University**

Master of Professional Studies, Applied Statistics (Data Science); GPA: 3.63/4.0

Ithaca, NY

Aug 2025 – Dec 2026
- **Amrita Vishwa Vidyapeetham**

B.Tech, Computer & Communication Engineering (Department Rank: III); GPA: 8.47/10

Coimbatore, India

Aug 2020 – May 2024

EXPERIENCE

- **Apple**

SiriAI UX Software Intern (Siri Home & System Experience)

Cupertino, CA

May 2026 – Aug 2026

- Built a team-configurable multi-agent pipeline automating bug routing, dedup, and prioritization — live in production across 2+ teams within Siri User Experiences, cutting 1 workday/week per engineer at lower operating cost than the incumbent tool.
 - Engineered modular, context-isolated agent specialization over monolithic prompting, driving 85% routing and 96% priority-bucketing accuracy.
 - Architected critic/debate agents for multi-agent conflict resolution, a vector-similarity search engine for duplicate detection, and automated root-cause log analysis, deployed on an internal AI platform via a shared dataset with per-team configuration.
- **Los Alamos National Laboratory**

Machine Learning Engineer (Industry Client Collaboration)

Los Alamos, NM

Jan 2026 – Present

- Designed a multi-model GenAI benchmarking pipeline quantifying probability calibration across GPT, Claude, Gemini, and Perplexity, producing calibration scorecards that guide when LLM outputs are trustworthy versus classical models.
 - Developed synthetic ground-truth generation via Gaussian/non-Gaussian mixture processes with exact Bayes posteriors, benchmarked against classical baselines (EM-GMM, k-means, random forests, gradient boosting) in a reproducible Python harness with automated MLOps tracking, scoring calibration via Brier score, ECE, reliability curves, and log loss.
- **PainMed PA**

Applied Machine Learning Engineer (Founder-Led Project)

Austin, TX

Sep 2025 – Present

- Delivered a human-in-the-loop ERA resubmission automation pipeline eliminating manual packet prep and shortening reimbursement turnaround — sustaining 10k+ claims/week while holding latency SLOs.
 - Built hybrid SQL and dual-vector-collection retrieval (payer guidelines vs. training manuals) with deterministic validation atop a fault-tolerant Python/SQL ETL into PostgreSQL on AWS (EC2, S3, RDS) and LangGraph/ChromaDB services with schema validators, retry/backoff, and Dockerized CI/CD — delivering auditable, hallucination-guarded checklists at 1.5s latency.
- **GE Healthcare**

Software Development Engineer (Edison Engineer)

Bengaluru, India

Aug 2024 – Jun 2025

- Deployed a product-routed RAG semantic search service automating hospital compliance questionnaire responses — cutting completion time from 60 to 20 minutes (67% faster) while keeping answers aligned with regulatory wording.
 - Optimized product-metadata filtering ahead of vector similarity search with hybrid cosine-plus-metadata ranking, eliminating cross-product answer leakage to deliver consistently accurate, precision-ranked suggestions.
 - Engineered the service in Python/FastAPI using HuggingFace encoders and ChromaDB on AWS (EC2, S3), with Dockerized CI/CD deployment and latency-optimized retrieval and inference paths.

PROJECTS

- **SignStream — Interactive ASL Recognition System:**

Architected and deployed a real-time deep learning inference service (Python, PyTorch, TensorFlow) with feature engineering and error analysis on MediaPipe landmarks, containerized via Docker on Linux with CI/CD, achieving 92.3% accuracy and <150ms end-to-end latency through optimized model training, evaluation, and deployment pipelines.

PUBLICATIONS

- **Optimizing ML Pipelines: Scalable Mobile-Based Tamil Sign Language Recognition:**

IEEE NKCon 2024

• **Smart Detection of Cyber Threats: A Machine Learning Approach:**

IJSREM 2024

CERTIFICATIONS

- **Certified ML Engineer Associate:**

Amazon Web Services

• **Machine Learning Specialization:**

Stanford (Coursera)

• **Generative AI for Data Scientists:**

IBM (Coursera)

SKILLS

- **Programming & Frameworks:**

Python, SQL, PyTorch, TensorFlow, scikit-learn, LangChain, Claude Code

• **Data & Systems:**

NumPy, Pandas, PostgreSQL, MongoDB, ChromaDB, REST APIs, Docker, Linux, CI/CD

• **Cloud:**

AWS (EC2, S3, RDS)

• **Competencies:**

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic Workflows, Model Evaluation, Feature Engineering, Unsupervised Learning, Statistical Modeling, ETL Pipelines, MLOps